

Practice Beginning Algebra Test IV (Version 1)

Name
Date
Course

Instructions

- i) As always, write a title, name, date, and course.
- ii) Number each problem, circle the number, and show work, even when you can do the problems in your head.
- iii) Draw a rectangle around your answers.
- iv) Staple a copy of these test questions with your work and your final answers.

① Identify the form of each linear equation below by writing "S" for Standard, "SI" for slope-intercept, or "PS" for Point-Slope.

a) $-3(x-3)=y-3$

PS

b) $\frac{596}{347} = 36.8x - 9y$

S

c) $y = \frac{7}{4}x + \frac{3}{16}$

SI

d) $x + y = 0$

S

② There are ducks and geese on a pond. The difference between the number of ducks and the number of geese is 21. The number of ducks is two more than twice the number of geese. How many geese are on the pond?

G = # of geese
D = # of ducks

$2 + 2G - G = 21$

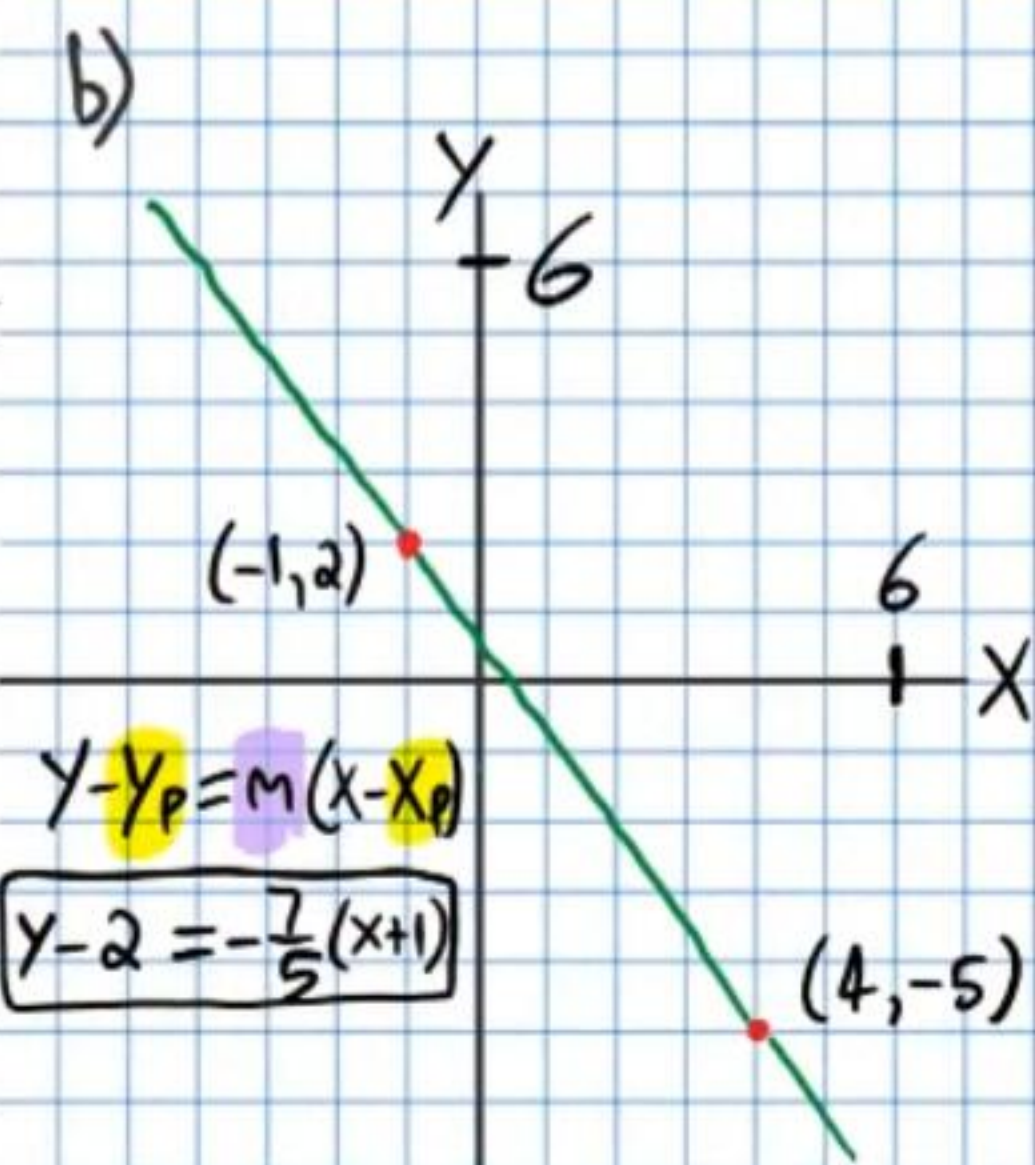
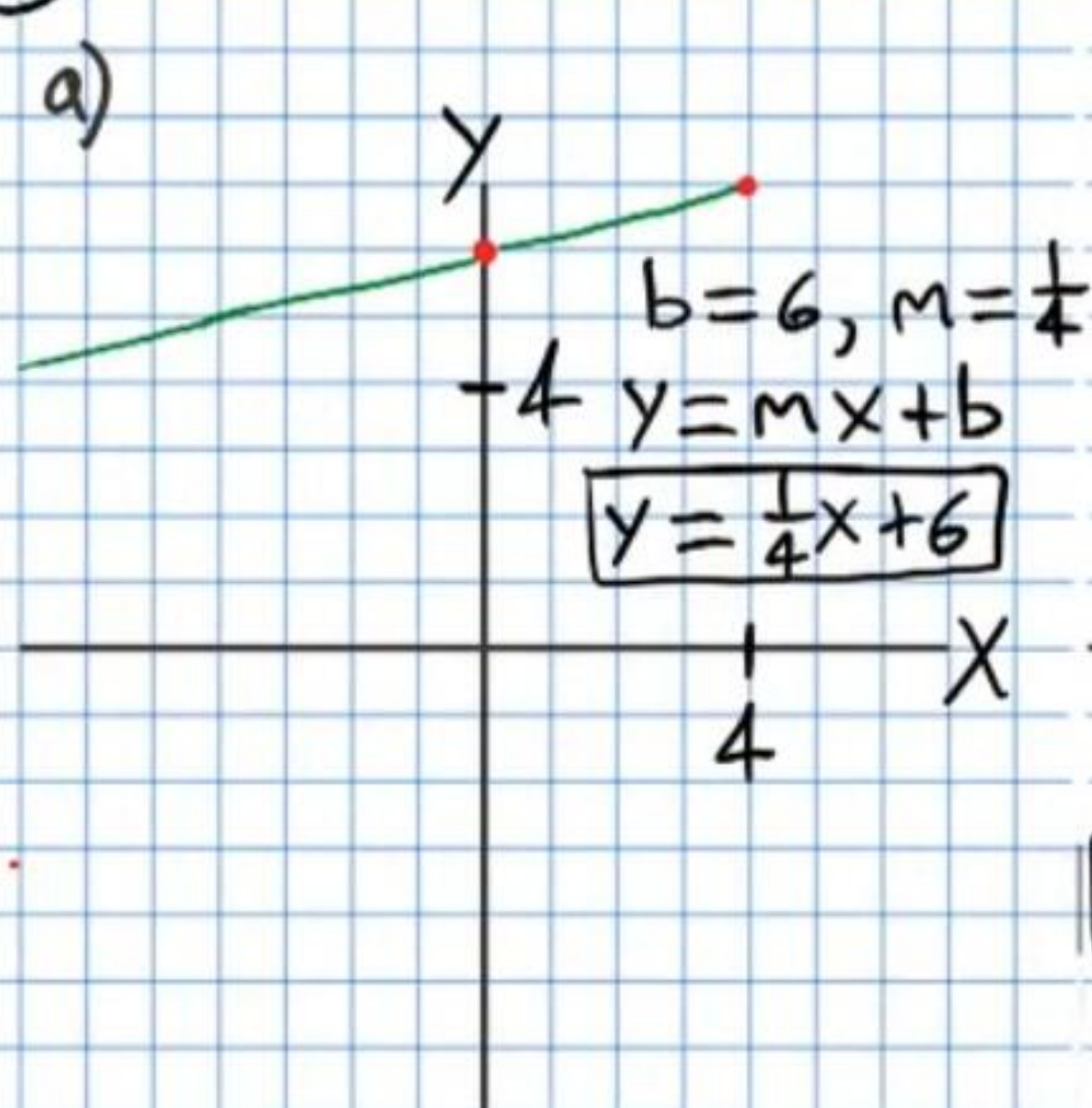
$2 + G = 21$

$G = 19$

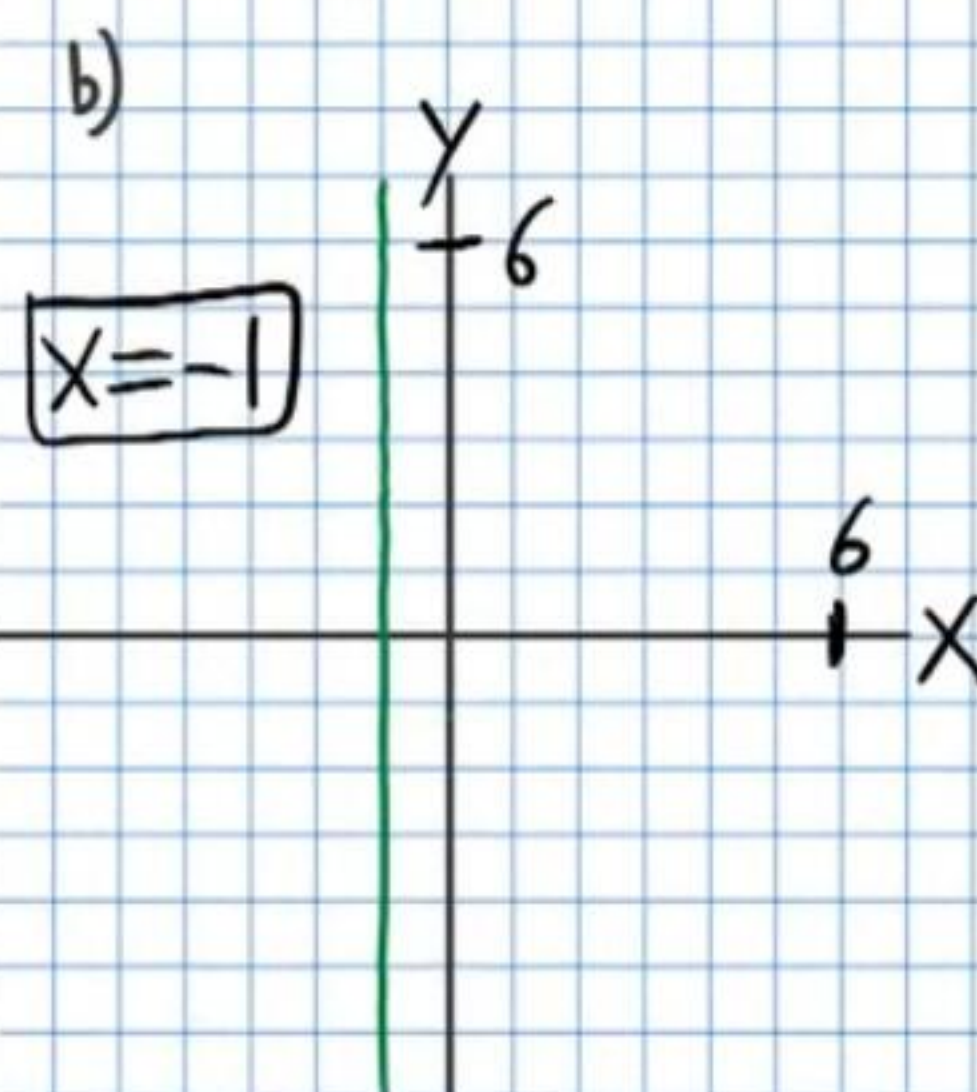
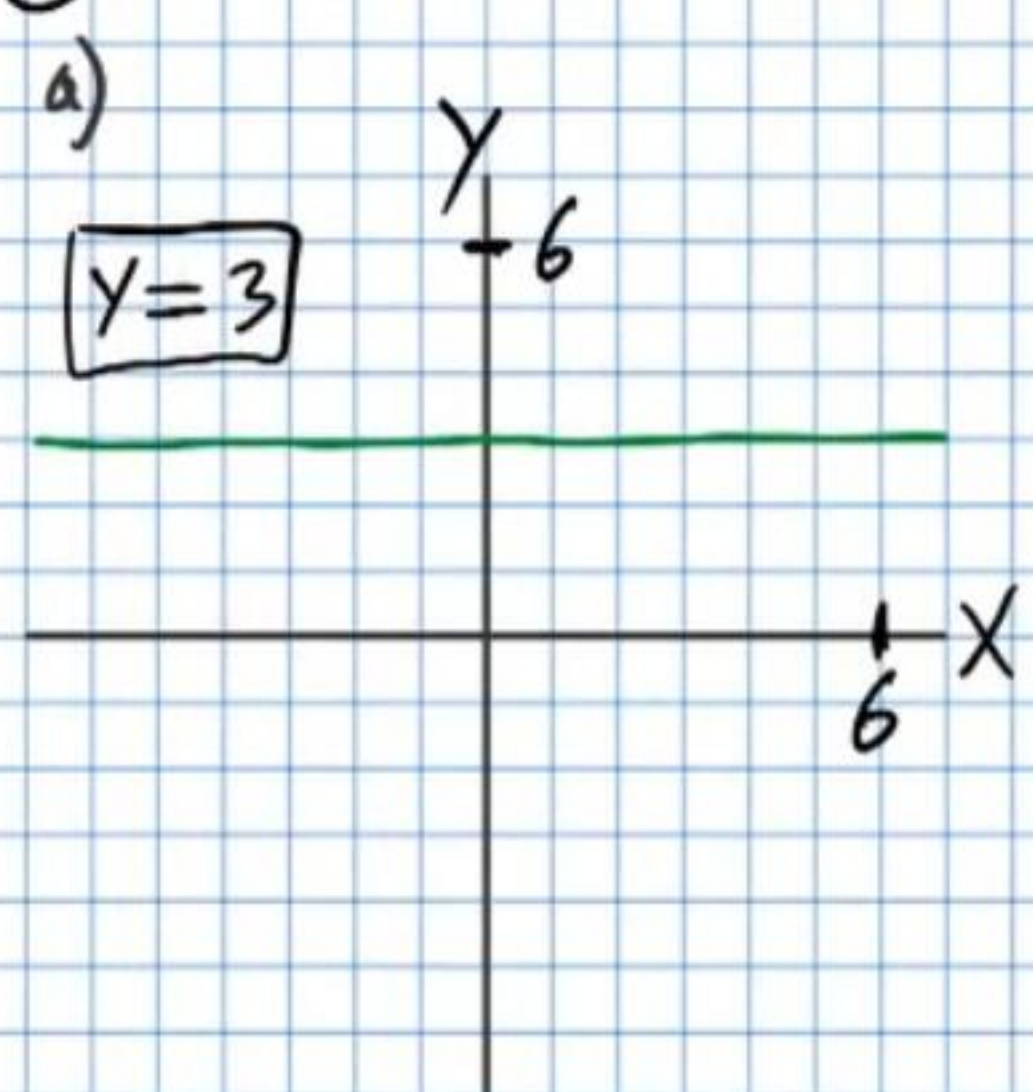
19 geese

For each graph below, write an equation of the line shown.

③



④



⑤ For each problem below, write an equation of the line with the given characteristics. There is more than one way to write the answers, but try to use the method that is quickest.

a) x-intercept: 6
slope: 4

$$(6, 0), m = 4$$

$$y - y_p = m(x - x_p)$$

$$y - 0 = 4(x - 6)$$

b) x-intercept: 3
y-intercept: 9

$$(3, 0) \text{ \& } (0, 9)$$

$$(x_1, y_1) \text{ \& } (x_2, y_2)$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{9 - 0}{0 - 3} = \frac{9}{-3} = -3$$

Method I: $m = -3$, $b = 9$

$$y = mx + b$$

$$y = -3x + 9$$

Method II: $(3, 0)$, $m = -3$

$$y - y_p = m(x - x_p)$$

$$y - 0 = -3(x - 3)$$

c) slope: $-\frac{1}{8}$
y-intercept: -11

$$y = mx + b$$

$$y = -\frac{1}{8}x - 11$$

d) slope: -4
point: $(0.7, -1.2)$

$$y - y_p = m(x - x_p)$$

$$y + 1.2 = -4(x - 0.7)$$

Solve.

⑥ $\begin{cases} 3x = y \\ -2x + 4y = 1 \end{cases}$

$$-2x + 4(3x) = 1$$

$$-2x + 12x = 1$$

$$10x = 1$$

$$x = \frac{1}{10}$$

$$3\left(\frac{1}{10}\right) = y$$

$$\frac{3}{10} = y$$

⑧ $\begin{cases} 4x + 5y = -12 \\ -3x - 10y = 9 \end{cases}$

$$2 \cdot \begin{cases} 4x + 5y = -12 \\ -3x - 10y = 9 \end{cases}$$

$$\begin{cases} 8x + 10y = -24 \\ -3x - 10y = 9 \end{cases}$$

$$5x + 0 = -15$$

$$x = -3$$

$$4(-3) + 5y = -12$$

$$y = 0$$

⑦ $\begin{cases} 10x + 7y = 5 \\ 4x + 7y = 23 \end{cases}$

$$-1 \cdot \begin{cases} 10x + 7y = 5 \\ 4x + 7y = 23 \end{cases}$$

$$\begin{cases} 10x + 7y = 5 \\ -4x - 7y = -23 \end{cases}$$

$$6x + 0 = -18$$

$$6x = -18$$

$$x = -3$$

$$4(-3) + 7y = 23$$

$$y = 5$$

⑨ $\begin{cases} x = 10 + 2y \\ 4y - x = -8 \end{cases}$

$$4y - (10 + 2y) = -8$$

$$4y - 10 - 2y = -8$$

$$2y - 10 = -8$$

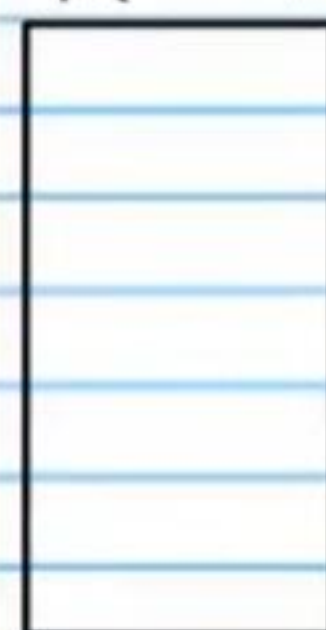
$$y = 1$$

$$x = 10 + 2(1)$$

$$x = 12$$

- ⑩ The perimeter of the rectangle below is 50 meters. Find the length of each side of the rectangle.

$$x^2 - 1$$



$$4x + 5$$

$$P = S_1 + S_2 + S_3 + S_4$$

$$50 = x^2 - 1 + x^2 - 1 + 4x + 5 + 4x + 5$$

$$50 = 2x^2 + 8x + 8$$

$$0 = 2x^2 + 8x - 42$$

$$\frac{0}{2} = \frac{2x^2 + 8x - 42}{2}$$

$$0 = x^2 + 4x - 21$$

$$0 = (x+7)(x-3)$$

$$x+7=0 \text{ or } x-3=0$$

$$x \neq -7 \text{ or } x=3$$

$$W = 3^2 - 1 = 8$$

$$L = 4(3) + 5 = 17$$

$$8\text{m}, 8\text{m}, 17\text{m}, 17\text{m}$$

- ⑪ Leonard sold 8 adult tickets and 7 child tickets for a total of \$132. Bella sold 5 adult tickets and 4 child tickets for a total of \$81. What is the cost per adult ticket and the cost per child ticket?

A = cost per adult ticket
C = cost per child ticket

$$8A + 7C = 132$$

$$5A + 4C = 81$$

$$-5 \cdot \begin{cases} 8A + 7C = 132 \\ 5A + 4C = 81 \end{cases}$$

$$8 \cdot \begin{cases} 8A + 7C = 132 \\ 5A + 4C = 81 \end{cases}$$

$$5 - 40A - 35C = -660$$

$$40A + 32C = 648$$

$$0 - 3C = -12$$

$$C = \$4/\text{child ticket}$$

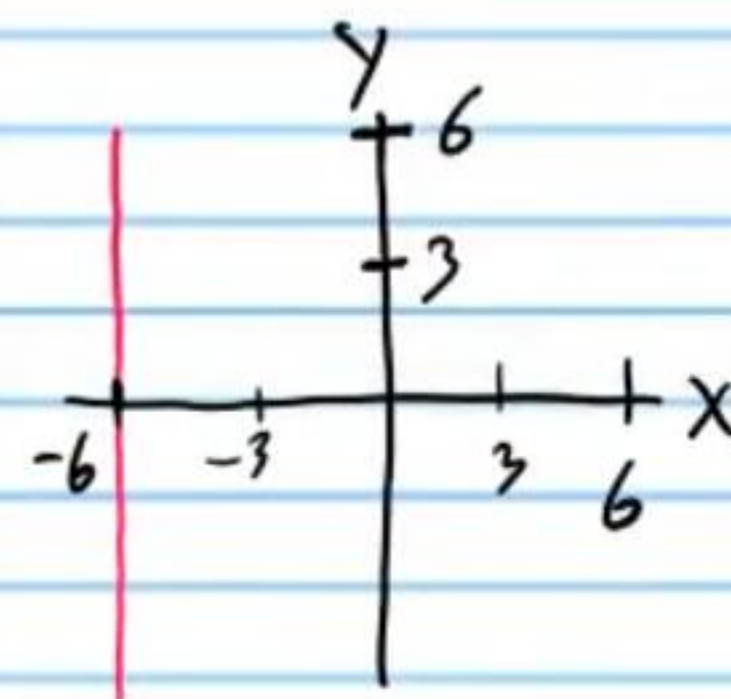
$$5A + 4(4) = 81$$

$$A = \$13/\text{Adult ticket}$$

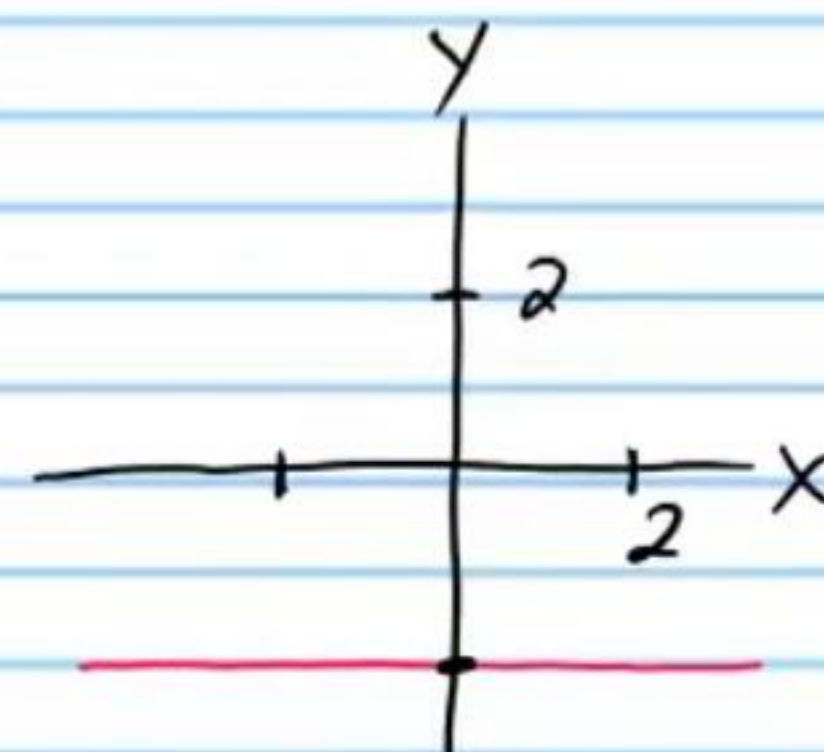
For problem #'s 12-15 below, always label the axes of your graphs and the intervals on each axis. Write all coordinates you use to make your graphs (e.g. (2,3)).

- ⑫ Graph the following equations.

a) $x = -6$



b) $y = -2$



- ⑬ a) Graph the following equation by plotting two points and connecting the dots (Method I).

$$x = 2 - y$$

Point I

$$y = 0$$

$$x = 2 - 0$$

$$x = 2$$

$$(2, 0)$$

Point II

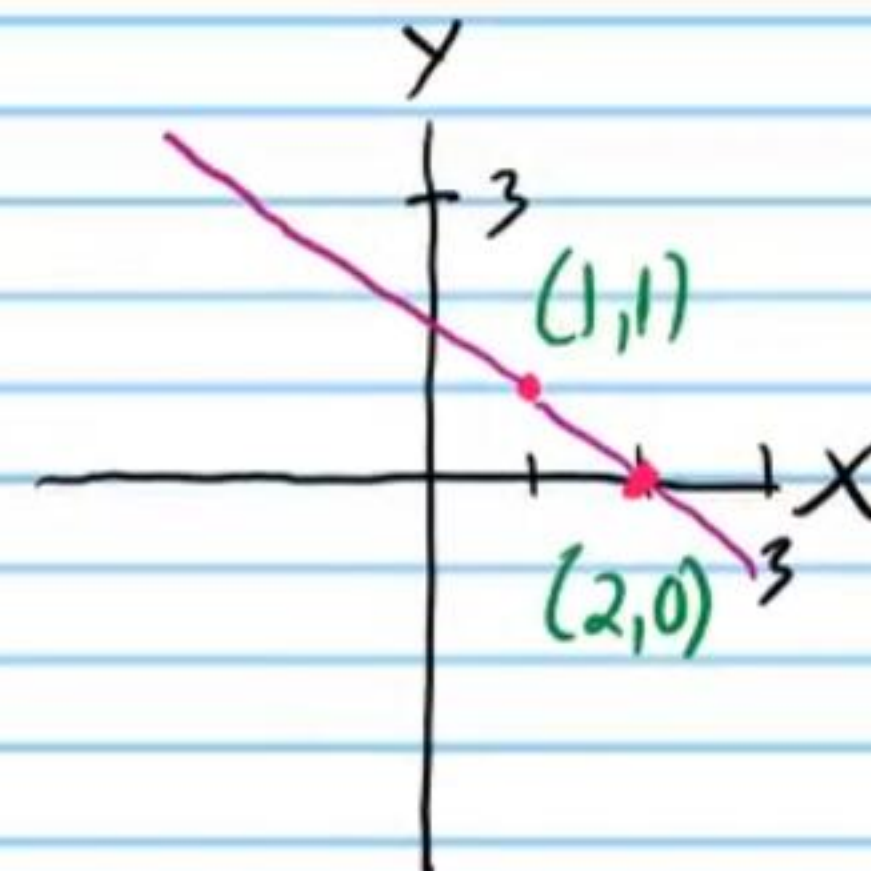
$$y = 1$$

$$x = 2 - 1$$

$$x = 1$$

$$(1, 1)$$

X	Y
2	0
1	1



b) Graph the following equation by using x and y intercepts (Method II).

$$6 = 2x + 3y$$

Point I

$$y = 0$$

$$6 = 2x + 3(0)$$

$$x = 3$$

$$(3, 0)$$

Point II

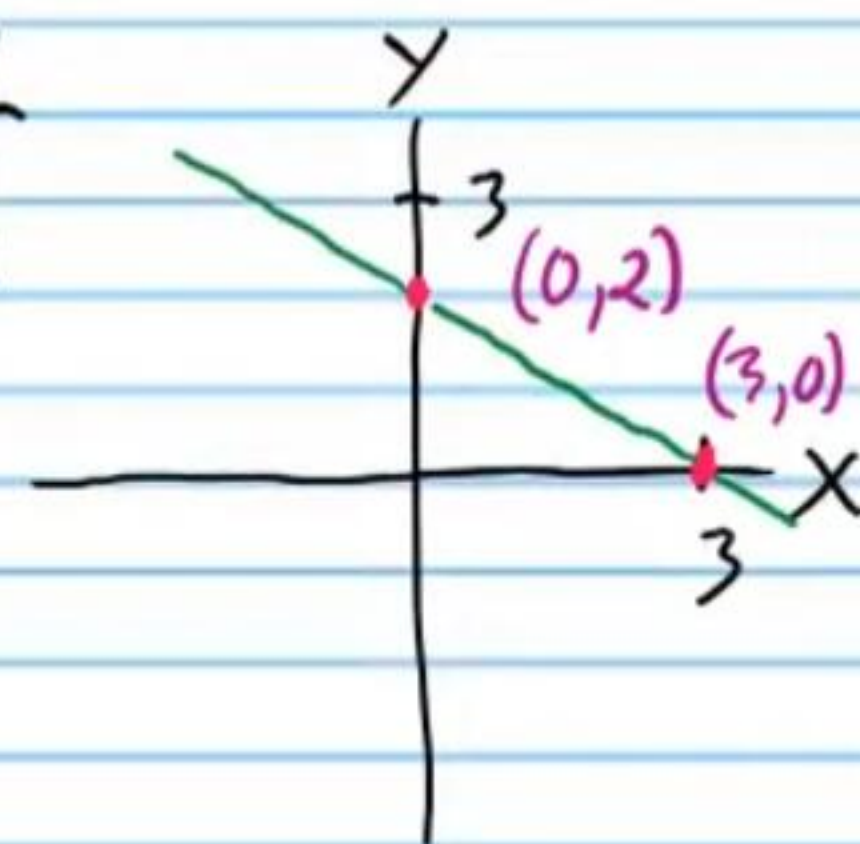
$$x = 0$$

$$6 = 2(0) + 3y$$

$$2 = y$$

$$(0, 2)$$

$$\begin{array}{c|c} x & y \\ \hline 3 & 0 \\ 0 & 2 \end{array}$$

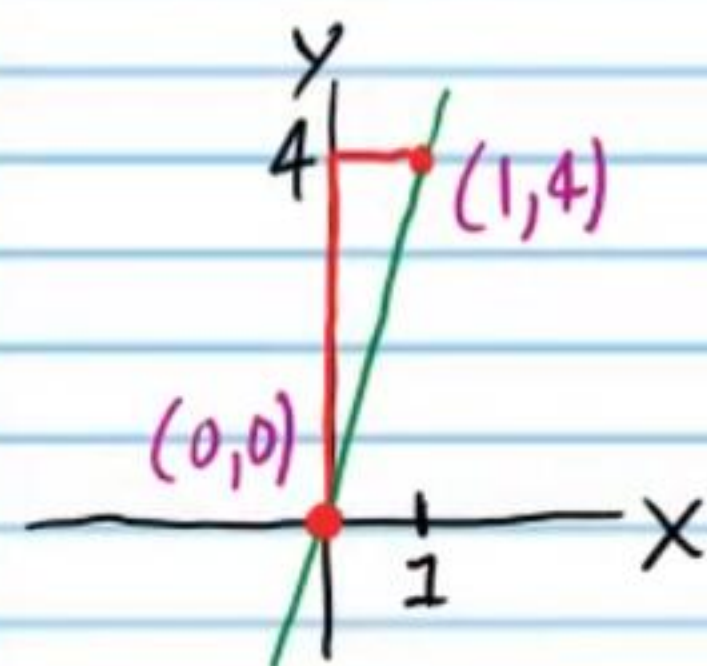


⑭ Graph the following equations by converting to slope-intercept form (if not already in this form) and using the y-intercept as a starting point and the slope to draw a triangle to find a second point (Method III).

a) $y = 4x$

$$y = mx + b$$

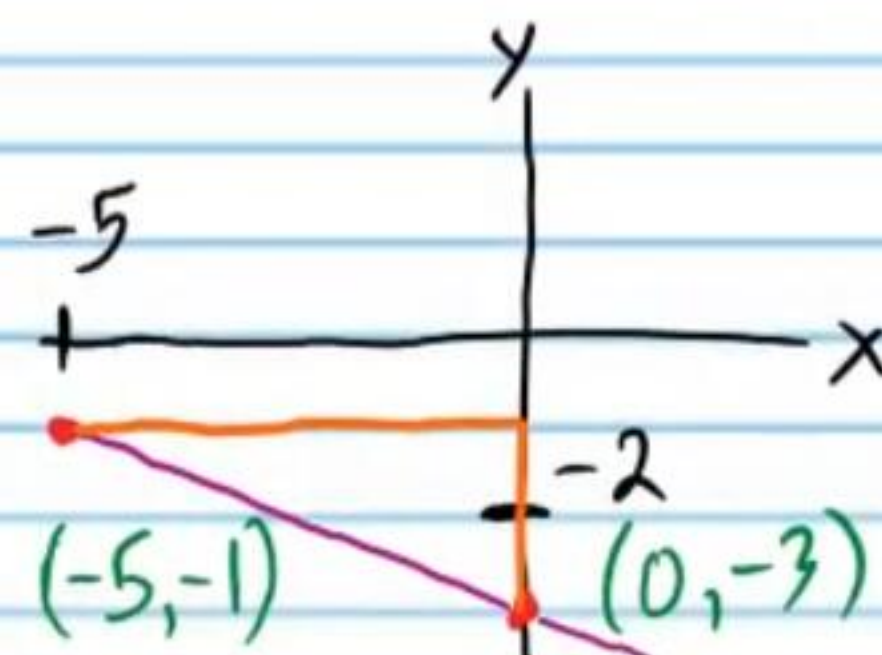
$$m = 4, b = 0$$



b) $-\frac{2}{5}x - 3 = y$

$$mx + b = y$$

$$m = -\frac{2}{5}, b = -3$$



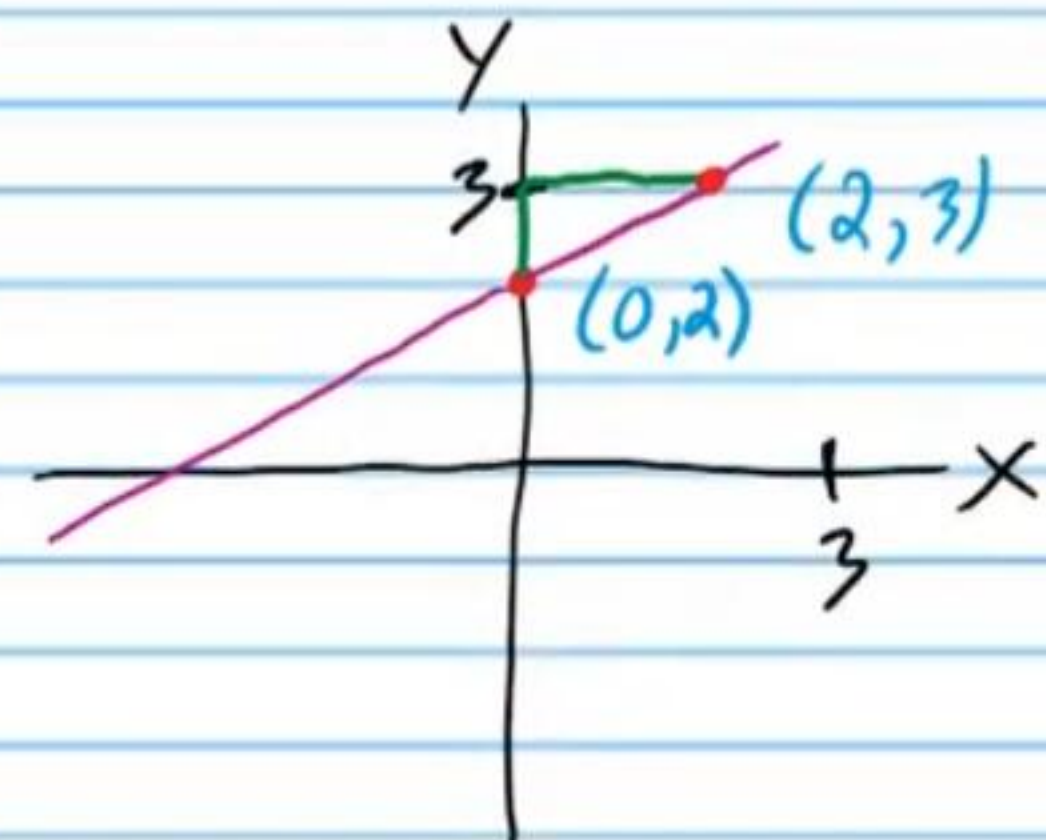
⑮ a) Graph the following equation by converting to slope-intercept form (if not already in this form) and using the y-intercept as a starting point and the slope to draw a triangle to find a second point (Method III).

$$y = \frac{x}{2} + 2$$

$$y = \frac{1}{2}x + 2$$

$$y = mx + b$$

$$m = \frac{1}{2}, b = 2$$

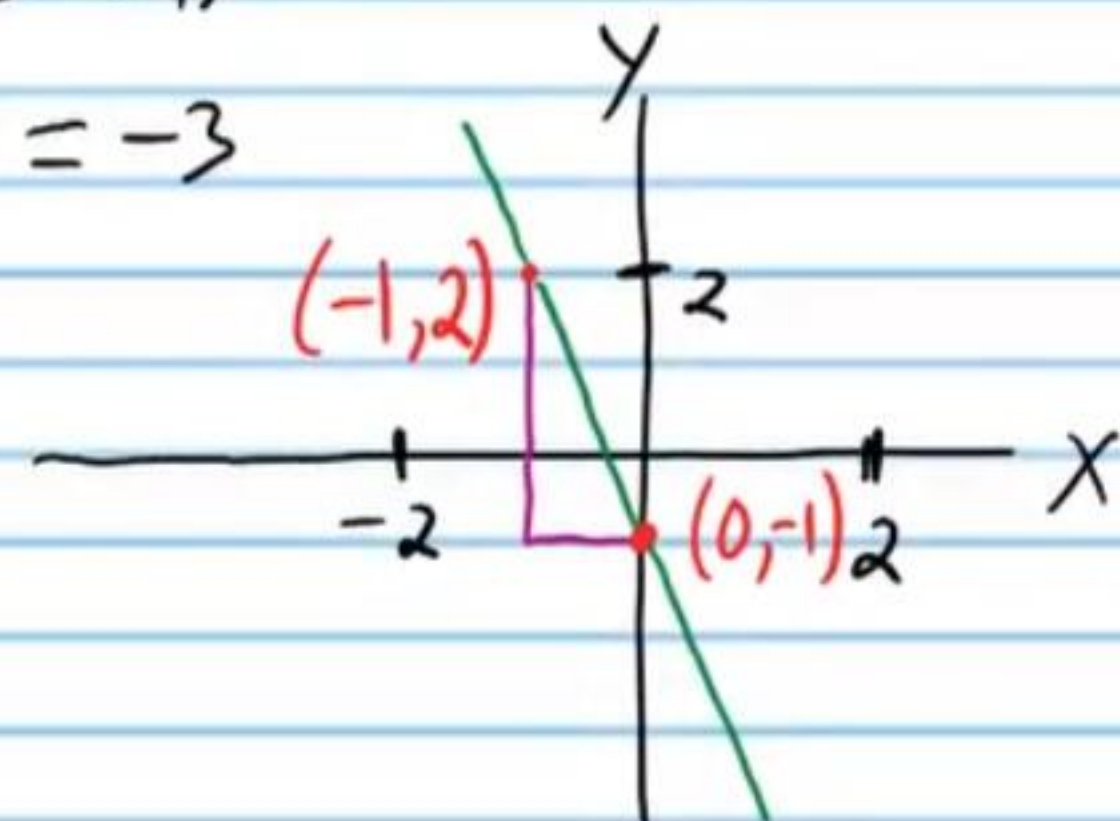


b) Graph the equation below by using its slope and a point on the line that is visible in the equation (Method IV).

$$y - 2 = -3(x + 1)$$

$$y - y_p = m(x - x_p)$$

$$(-1, 2), m = -3$$



- ⑩ a) Use the slope equation to find the slope of the line containing the two points below.
 b) Then write a point-slope form of the equation of the line containing the two points below.
 c) Then convert the equation from point-slope form to slope-intercept form.
 d) Find the y-intercept of the line.

$$(-4, 7) \text{ \& } (3, -2)$$

$$(x_1, y_1) \text{ \& } (x_2, y_2)$$

$$b) y - y_p = m(x - x_p)$$

$$d) b = \frac{13}{7}$$

$$y - 7 = -\frac{9}{7}(x + 4)$$

$$a) m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{-2 - 7}{3 - (-4)} = \boxed{-\frac{9}{7}}$$

$$c) y - 7 = -\frac{9}{7}(x + 4)$$

$$y - 7 = -\frac{9}{7}x - \frac{36}{7}$$

$$y = -\frac{9}{7}x + \frac{13}{7}$$

- ⑪ Sour dough bread costs \$3.18 per loaf and rye bread costs \$2.29 per loaf. Ellis spent \$68.18 on these two items. He bought 10 times more sour dough loaves than rye loaves. How many loaves of each type of bread did he buy?

$$S = \# \text{ of loaves of sour}$$

$$R = \# \text{ of loaves of rye}$$

$$34.09R = 68.18$$

$$R = 2 \text{ rye loaves}$$

$$\begin{cases} S = 10R \\ 3.18S + 2.29R = 68.18 \end{cases}$$

$$\rightarrow 3.18(10R) + 2.29R = 68.18$$

$$31.8R + 2.29R = 68.18$$

$$S = 10(2)$$

$$S = 20 \text{ sour loaves}$$

- ⑫ Use the slope-intercept equation to find the slope-intercept form of the line with the given characteristics.

- a) Y-intercept = 2, contains the point (-1, -7).

$$b = 2$$

$$y = mx + b$$

$$-7 = m(-1) + 2$$

$$-7 = -m + 2$$

$$m = 9$$

$$m = 9, b = 2$$

$$y = mx + b$$

$$y = 9x + 2$$

- b) Slope = 4, contains the point (-5, -3).

$$m = 4$$

$$y = mx + b$$

$$-3 = 4(-5) + b$$

$$b = 17, m = 4$$

$$y = mx + b$$

$$y = 4x + 17$$

- ⑬ For each problem below, write an equation of the line with the given characteristics.

- a) Parallel to $y = -2x + 3.8$ and containing the point (5, 2).

$$m_{||} = -2, (5, 2)$$

$$y - y_p = m(x - x_p)$$

$$y - 2 = -2(x - 5)$$

- b) Parallel to $y + 9 = \frac{2}{3}(x - 1)$ and containing the point (-6, 3).

$$m_{||} = \frac{2}{3}, (-6, 3)$$

$$y - y_p = m(x - x_p)$$

$$y - 3 = \frac{2}{3}(x + 6)$$

c) Perpendicular to $76(x+96)=y+32$ and containing the point $(1, -1)$.

$$M_{\perp} = -\frac{1}{76}, (1, -1) \quad \begin{array}{l} m(x-x_p) = y-y_p \\ y-y_p = m(x-x_p) \\ y+1 = -\frac{1}{76}(x-1) \end{array}$$

d) Perpendicular to $8x+5y=-5$ and containing the point $(3, 1)$.

$$\begin{array}{l} 8x+5y=-5 \\ 5y=-8x-5 \\ y=-\frac{8}{5}x-1 \end{array} \quad \begin{array}{l} M_{\perp} = \frac{5}{8} \\ y-y_p = m(x-x_p) \\ y-1 = \frac{5}{8}(x-3) \end{array}$$

20) Daniel has dimes and quarters in his pocket. The value of these coins is \$4.85. If he has a total of 26 coins, how many of each type of coin does he have?

D = # of dimes
Q = # of quarters

$$\begin{cases} D+Q=26 \\ 0.1D+0.25Q=4.85 \end{cases}$$

$$\begin{array}{l} -0.1 \cdot \begin{cases} D+Q=26 \\ 0.1D+0.25Q=4.85 \end{cases} \\ \hline \begin{cases} -0.1D-0.1Q=-2.6 \\ 0.1D+0.25Q=4.85 \end{cases} \end{array}$$

$$\begin{array}{l} -0.1D-0.1Q=-2.6 \\ 0.1D+0.25Q=4.85 \\ \hline 0+0.15Q=2.25 \end{array}$$

$$0 + 0.15Q = 2.25$$

$$Q = 15 \text{ quarters}$$

$$D + 15 = 26$$

$$D = 11 \text{ dimes}$$

21) Chris wants to make 3.6 gallons of fuel that is 20% alcohol. He has fuel that is 30% alcohol and fuel that is 14% alcohol. How many gallons of each mixture should he combine?

X = gallons of 30%

y = gallons of 14%

$$\begin{cases} X+y=3.6 \\ 0.3X+0.14y=0.2(3.6) \end{cases}$$

$$\begin{array}{l} -0.3 \cdot \begin{cases} X+y=3.6 \\ 0.3X+0.14y=0.72 \end{cases} \\ \hline \begin{cases} -0.3X-0.3y=-1.08 \\ 0.3X+0.14y=0.72 \end{cases} \end{array}$$

$$\begin{array}{l} -0.3X-0.3y=-1.08 \\ 0.3X+0.14y=0.72 \\ \hline 0+0.16y=-0.36 \end{array}$$

$$0 + 0.16y = -0.36$$

$$y = 2.25 \text{ gal } 14\%$$

$$X + 2.25 = 3.6$$

$$X = 1.35 \text{ gal } 30\%$$

* You must use graphing paper for problem #16.

22 a) Solve the system of equations below using the graphing method. As usual, label axes, intervals, and any points you use to solve the system. You might not be able to solve the system using this method, so move on to part (b) to help you label the coordinates of intersection in part (a).

$$\begin{cases} 9x - 8y = 4 \\ -6x + 7y = -2 \end{cases}$$

Line I

Point A

$$x = 0$$

$$9(0) - 8y = 4$$

$$y = -\frac{1}{2}$$

$$(0, -\frac{1}{2})$$

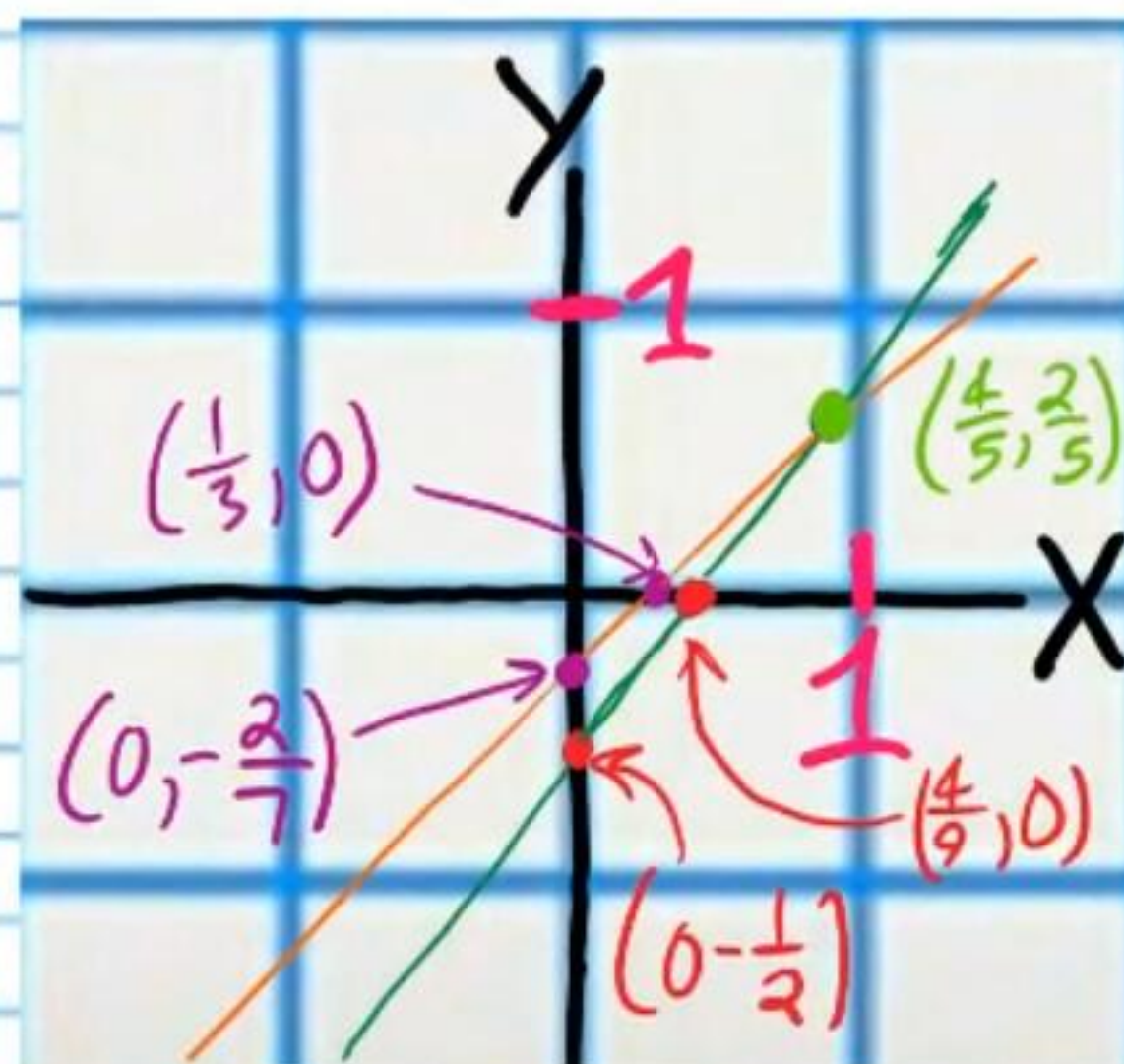
Point B

$$y = 0$$

$$9x - 8(0) = 4$$

$$x = \frac{4}{9}$$

$$(\frac{4}{9}, 0)$$



Line II

Point C

$$x = 0$$

$$-6(0) + 7y = -2$$

$$y = -\frac{2}{7}$$

$$(0, -\frac{2}{7})$$

Point D

$$y = 0$$

$$-6x + 7(0) = -2$$

$$x = \frac{1}{3}$$

$$(\frac{1}{3}, 0)$$

b) Then check your solution by solving the system algebraically (i.e. with substitution or elimination).

$$\begin{cases} 9x - 8y = 4 \\ -6x + 7y = -2 \end{cases}$$

$$\begin{cases} 18x - 16y = 8 \\ -18x + 21y = -6 \end{cases}$$

$$5y = 2$$

$$\boxed{y = \frac{2}{5}}$$

$$-6x + 7(\frac{2}{5}) = -2$$

$$-6x + \frac{14}{5} = -2$$

$$-6x = -\frac{24}{5}$$

$$x = -\frac{24}{5} \div (-6)$$

$$x = -\frac{24}{5} \cdot (-\frac{1}{6})$$

$$\boxed{x = \frac{4}{5}}$$

②③ Jeff wants to make 20 ounces of a mixture of jelly beans and gumdrops that costs \$3 per ounce. He has a mixture of jelly beans and gumdrops that costs \$1 per ounce and a mixture of jelly beans and gumdrops that costs \$7 per ounce. How many ounces of each mixture should he combine?

$x = \#$ of ounces of \$1/oz mixture
 $y = \#$ of ounces of \$7/oz mixture

$$\begin{cases} x + y = 20 \\ 1x + 7y = 20 \quad (3) \end{cases}$$

$$\begin{cases} x + y = 20 \\ x + 7y = 60 \end{cases}$$

$$\begin{cases} -1 \cdot x + y = 20 \\ x + 7y = 60 \end{cases}$$

$$\begin{cases} -x - y = -20 \\ x + 7y = 60 \end{cases}$$

$$6y = 40$$

$$y = 6\frac{2}{3} \text{ oz of } \$7/\text{oz}$$

$$x + 6\frac{2}{3} = 20$$

$$x = 13\frac{1}{3} \text{ oz } \$1/\text{oz}$$

②④ Bill is 6 feet tall. He slings a rock up into the air. The height of the rock in feet t seconds after release can be modeled by the following equation (neglecting air resistance).

$$h = -16t^2 + 95t + 6$$

a) Find the height of the rock two seconds after release.

$$h = -16(2)^2 + 95(2) + 6 = -16(4) + 190 + 6 = -64 + 196 = \boxed{132 \text{ ft}}$$

b) How many seconds after release will it take for the rock to reach a height of 85 feet?

$$85 = -16t^2 + 95t + 6$$

$$16t^2 - 95t + 79 = 0$$

$$(16t - 79)(t - 1) = 0$$

$$16t - 79 = 0 \text{ or } t - 1 = 0$$

$$t = \frac{79}{16} \text{ s or } t = 1 \text{ s}$$

c) What is the height of the rock at the exact moment that it is released from the sling?

$$h = -16t^2 + 95t + 6$$

$$h = -16(0)^2 + 95(0) + 6 = \boxed{6 \text{ ft}}$$

d) How long will the rock be in the air before it lands on the ground?

$$0 = -16t^2 + 95t + 6$$

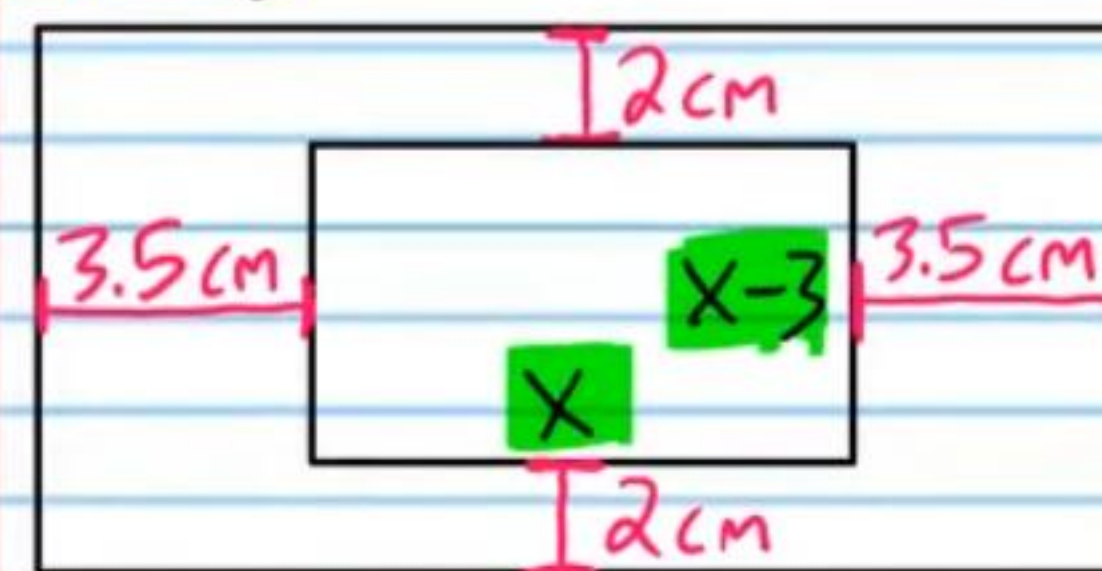
$$16t^2 - 95t - 6 = 0$$

$$(16t + 1)(t - 6) = 0$$

$$16t + 1 = 0 \text{ or } t - 6 = 0$$

$$t \neq -\frac{1}{16} \text{ or } \boxed{t = 6 \text{ s}}$$

(25) The top of a tissue box is rectangular. There is a smaller rectangular opening where the tissue comes out near the center of the top of the box. The area of the top of the tissue box is 40 square centimeters more than twice the area of the rectangular opening. The length of the opening is three centimeters less than the width of the opening. Find the dimensions of the top of the tissue box.



$$A_T = 2A_{RO} + 40$$

$$(x + 3.5 + 3.5)(x - 3 + 2 + 2) = 2[x(x - 3)] + 40$$

$$(x + 7)(x + 1) = 2x(x - 3) + 40$$

$$x^2 + 8x + 7 = 2x^2 - 6x + 40$$

$$0 = x^2 - 14x + 33$$

$$0 = (x - 11)(x - 3)$$

$$x - 11 = 0 \text{ or } x - 3 = 0$$

$$x = 11 \text{ or } x \neq 3$$

$$W = 11 + 3.5 + 3.5 = \boxed{18 \text{ cm}}, L = 11 - 3 + 2 + 2 = \boxed{12 \text{ cm}}$$